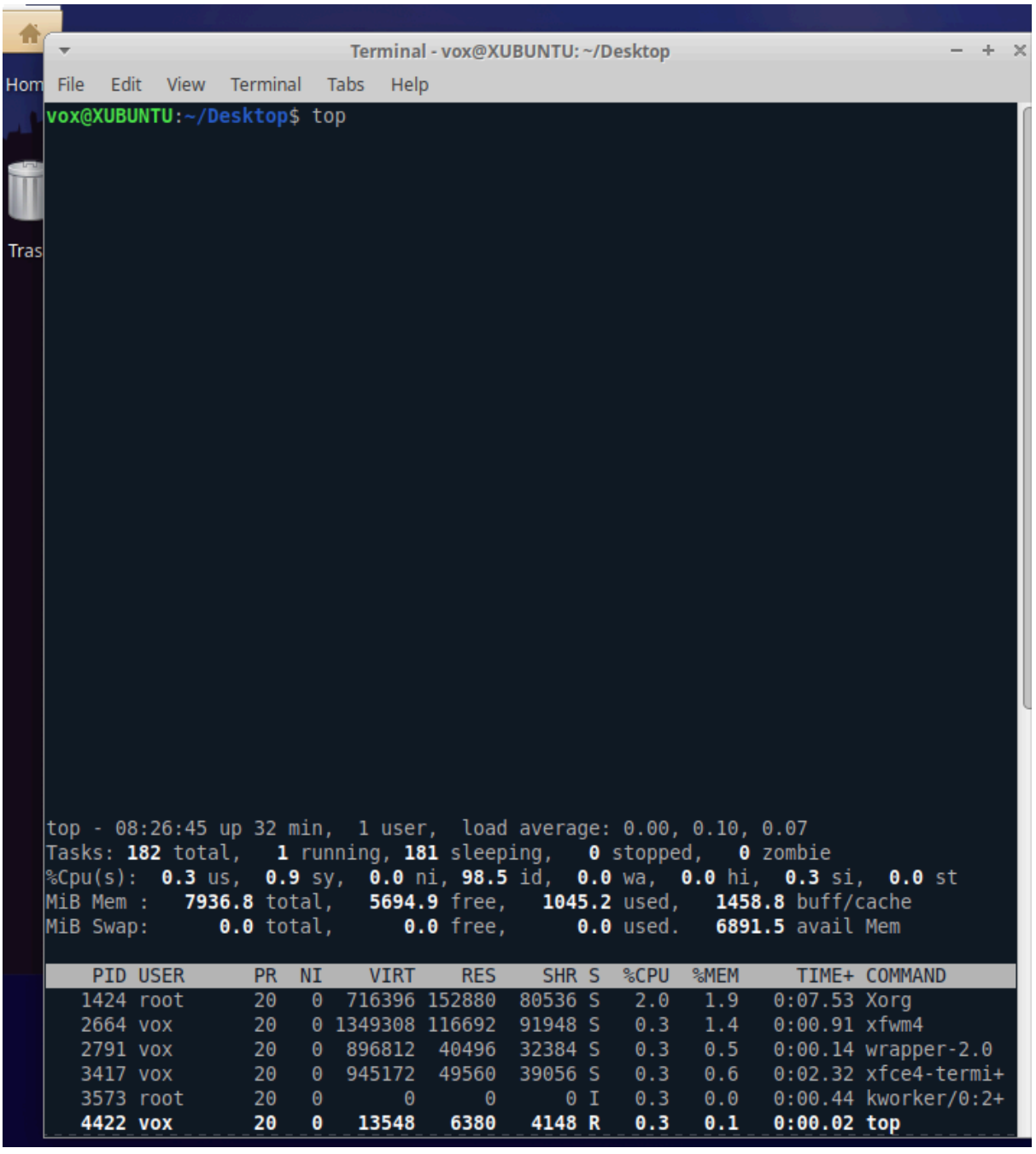
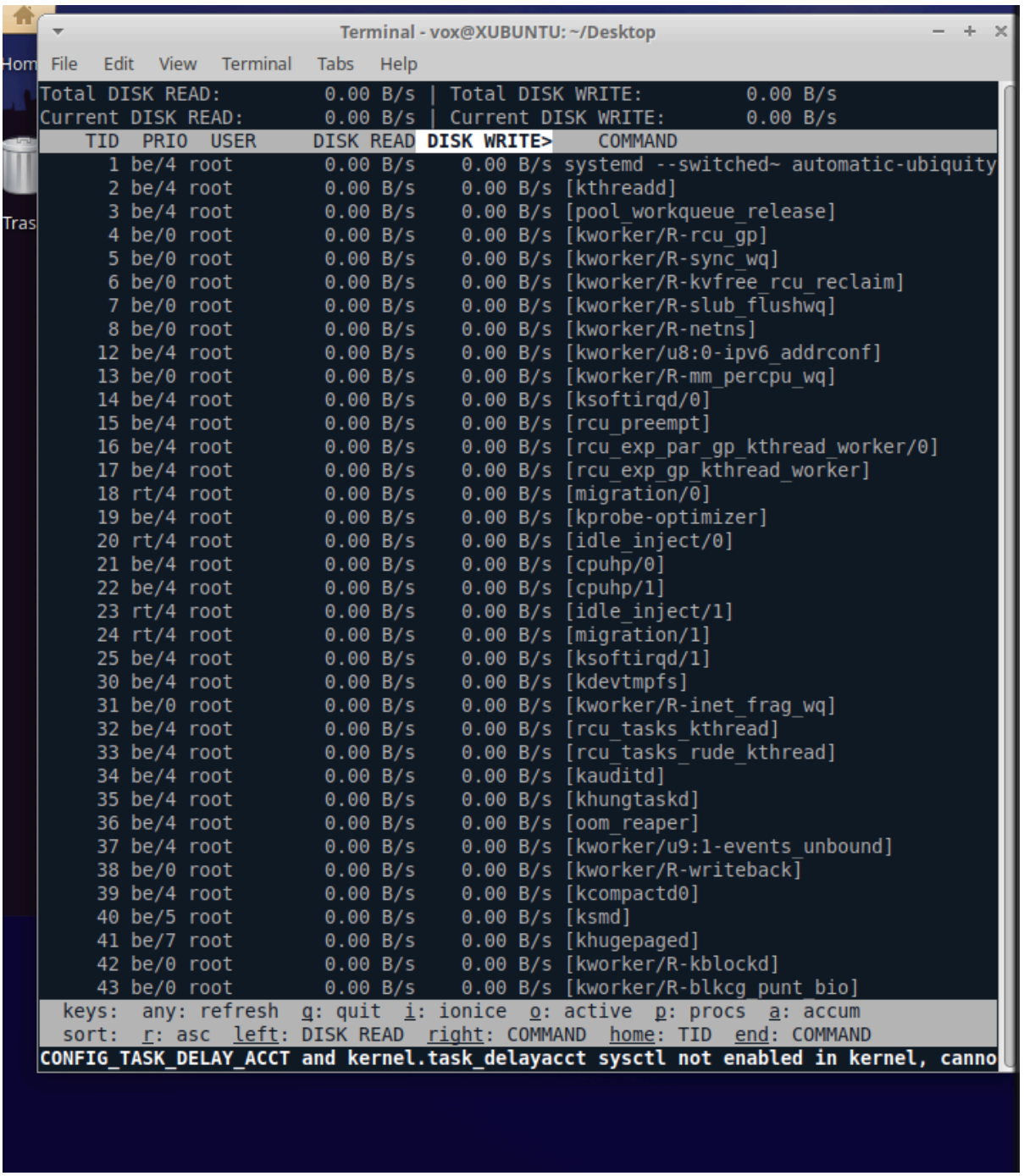
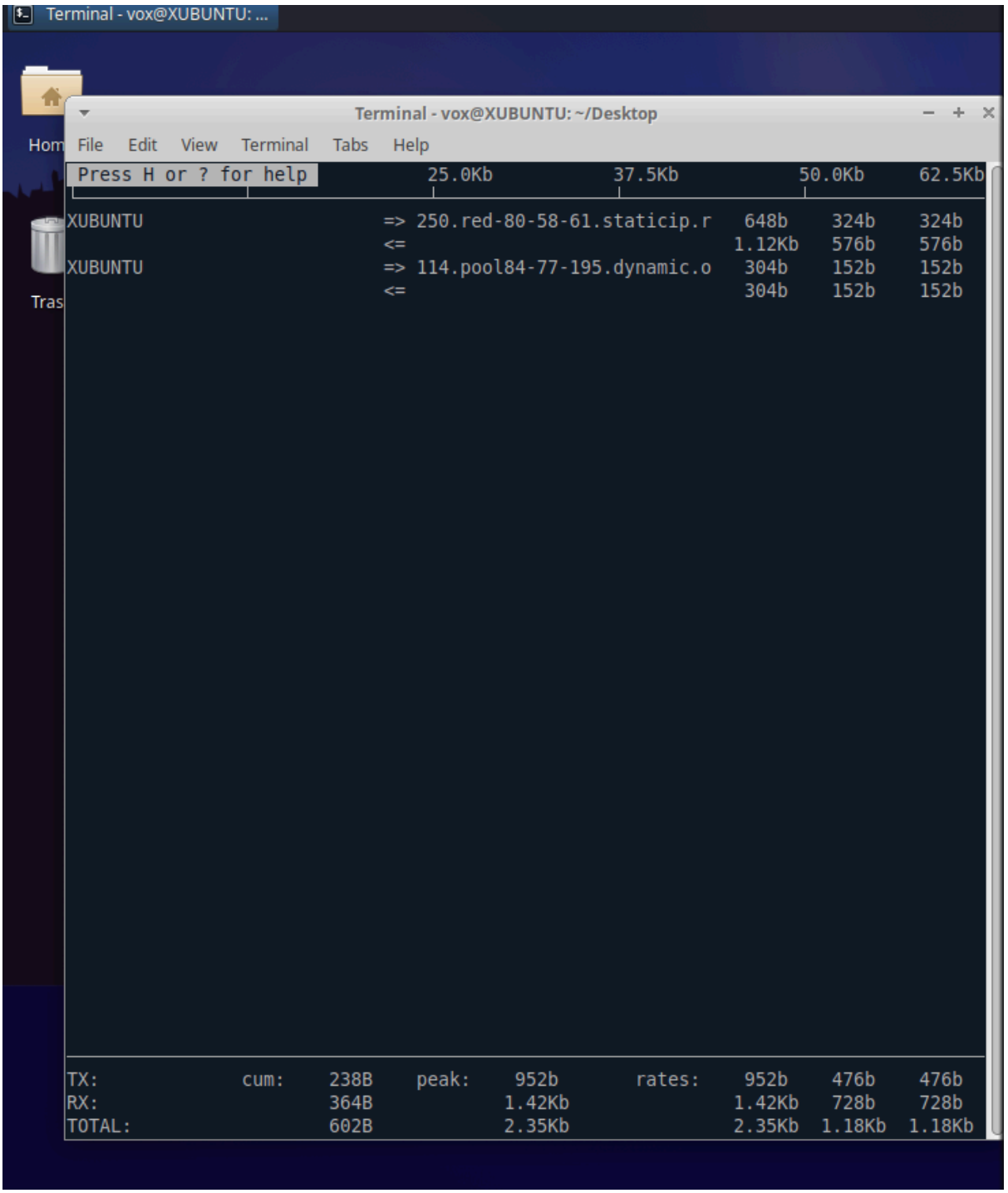
Monitorización, procesos y servicios
Realiza los siguientes ejercicios, de los que recibes una pista y cuál es una de sus utilidades principales en la administración de sistemas.

(F) Monitoriza CPU y RAM en tiempo real.
Pista: htop o top.
Utilidad: detectar consumo excesivo.

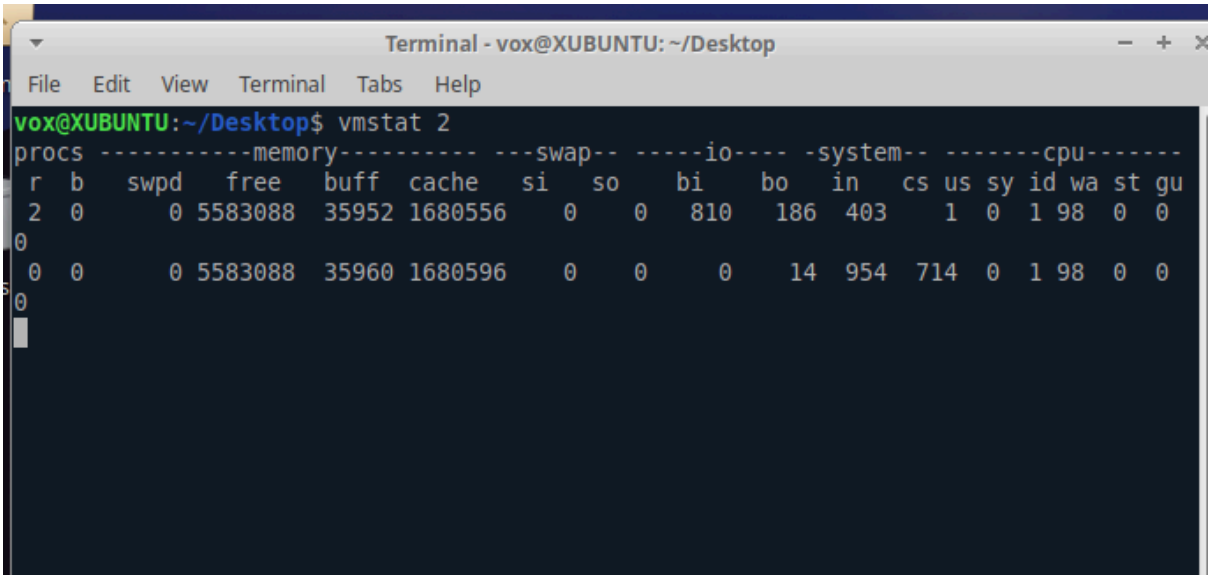


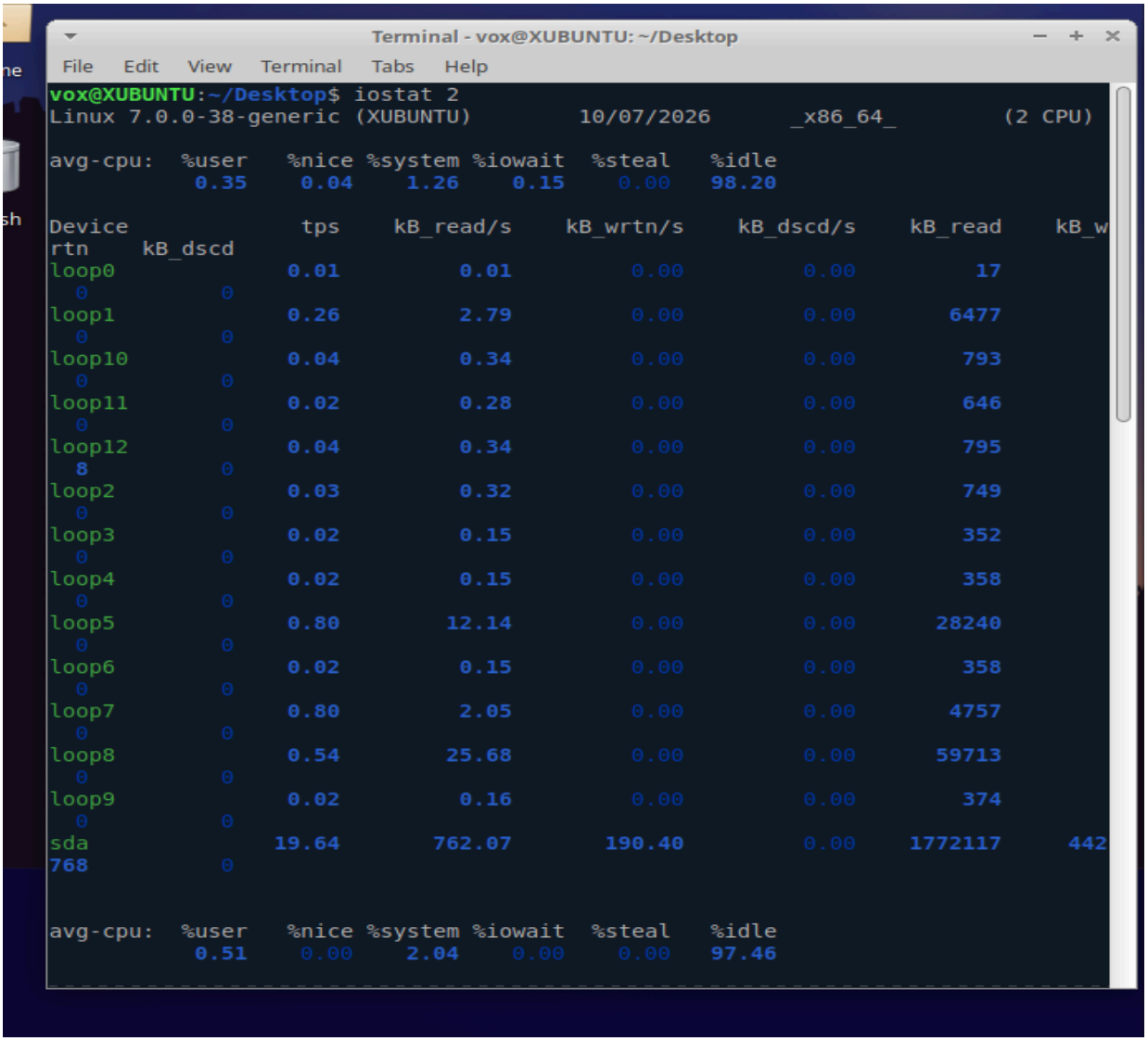
(F) Revisa E/S por proceso y ancho de banda de red.
Pista: iotop e iftop.
Utilidad: localizar cuellos de botella.


Terminal - vox@XUBUNTU: ~/Desktop
File Edit View Terminal Tabs Help
Total DISK READ: 0.00 B/s | Total DISK WRITE: 0.00 B/s
Current DISK READ: 0.00 B/s | Current DISK WRITE: 0.00 B/s
TID PRIO USER DISK READ DISK WRITE> COMMAND
1 be/4 root 0.00 B/s 0.00 B/s systemd --switched~ automatic-ubiquity
2 be/4 root 0.00 B/s 0.00 B/s [kthreadd]
3 be/4 root 0.00 B/s 0.00 B/s [pool_workqueue_release]
4 be/0 root 0.00 B/s 0.00 B/s [kworker/R-rcu_gp]
5 be/0 root 0.00 B/s 0.00 B/s [kworker/R-sync_wq]
6 be/0 root 0.00 B/s 0.00 B/s [kworker/R-kvfree_rcu_reclaim]
7 be/0 root 0.00 B/s 0.00 B/s [kworker/R-slub_flushwq]
8 be/0 root 0.00 B/s 0.00 B/s [kworker/R-netns]
12 be/4 root 0.00 B/s 0.00 B/s [kworker/u8:0-ipv6_addrconf]
13 be/0 root 0.00 B/s 0.00 B/s [kworker/R-mm_percpu_wq]
14 be/4 root 0.00 B/s 0.00 B/s [ksoftirqd/0]
15 be/4 root 0.00 B/s 0.00 B/s [rcu_preempt]
16 be/4 root 0.00 B/s 0.00 B/s [rcu_exp_par_gp_kthread_worker/0]
17 be/4 root 0.00 B/s 0.00 B/s [rcu_exp_gp_kthread_worker]
18 rt/4 root 0.00 B/s 0.00 B/s [migration/0]
19 be/4 root 0.00 B/s 0.00 B/s [kprobe-optimizer]
20 rt/4 root 0.00 B/s 0.00 B/s [idle_inject/0]
21 be/4 root 0.00 B/s 0.00 B/s [cpuhp/0]
22 be/4 root 0.00 B/s 0.00 B/s [cpuhp/1]
23 rt/4 root 0.00 B/s 0.00 B/s [idle_inject/1]
24 rt/4 root 0.00 B/s 0.00 B/s [migration/1]
25 be/4 root 0.00 B/s 0.00 B/s [ksoftirqd/1]
30 be/4 root 0.00 B/s 0.00 B/s [kdevtmpfs]
31 be/0 root 0.00 B/s 0.00 B/s [kworker/R-inet_frag_wq]
32 be/4 root 0.00 B/s 0.00 B/s [rcu_tasks_kthread]
33 be/4 root 0.00 B/s 0.00 B/s [rcu_tasks_rude_kthread]
34 be/4 root 0.00 B/s 0.00 B/s [kauditd]
35 be/4 root 0.00 B/s 0.00 B/s [khungtaskd]
36 be/4 root 0.00 B/s 0.00 B/s [oom_reaper]
37 be/4 root 0.00 B/s 0.00 B/s [kworker/u9:1-events_unbound]
38 be/0 root 0.00 B/s 0.00 B/s [kworker/R-writeback]
39 be/4 root 0.00 B/s 0.00 B/s [kcompactd0]
40 be/5 root 0.00 B/s 0.00 B/s [ksmd]
41 be/7 root 0.00 B/s 0.00 B/s [khugepaged]
42 be/0 root 0.00 B/s 0.00 B/s [kworker/R-kblockd]
43 be/0 root 0.00 B/s 0.00 B/s [kworker/R-blkcg_punt_bio]
keys: any: refresh q: quit i: ionice o: active p: procs a: accum
sort: r: asc left: DISK READ right: COMMAND home: TID end: COMMAND
CONFIG_TASK_DELAY_ACCT and kernel.task_delayacct sysctl not enabled in kernel, canno

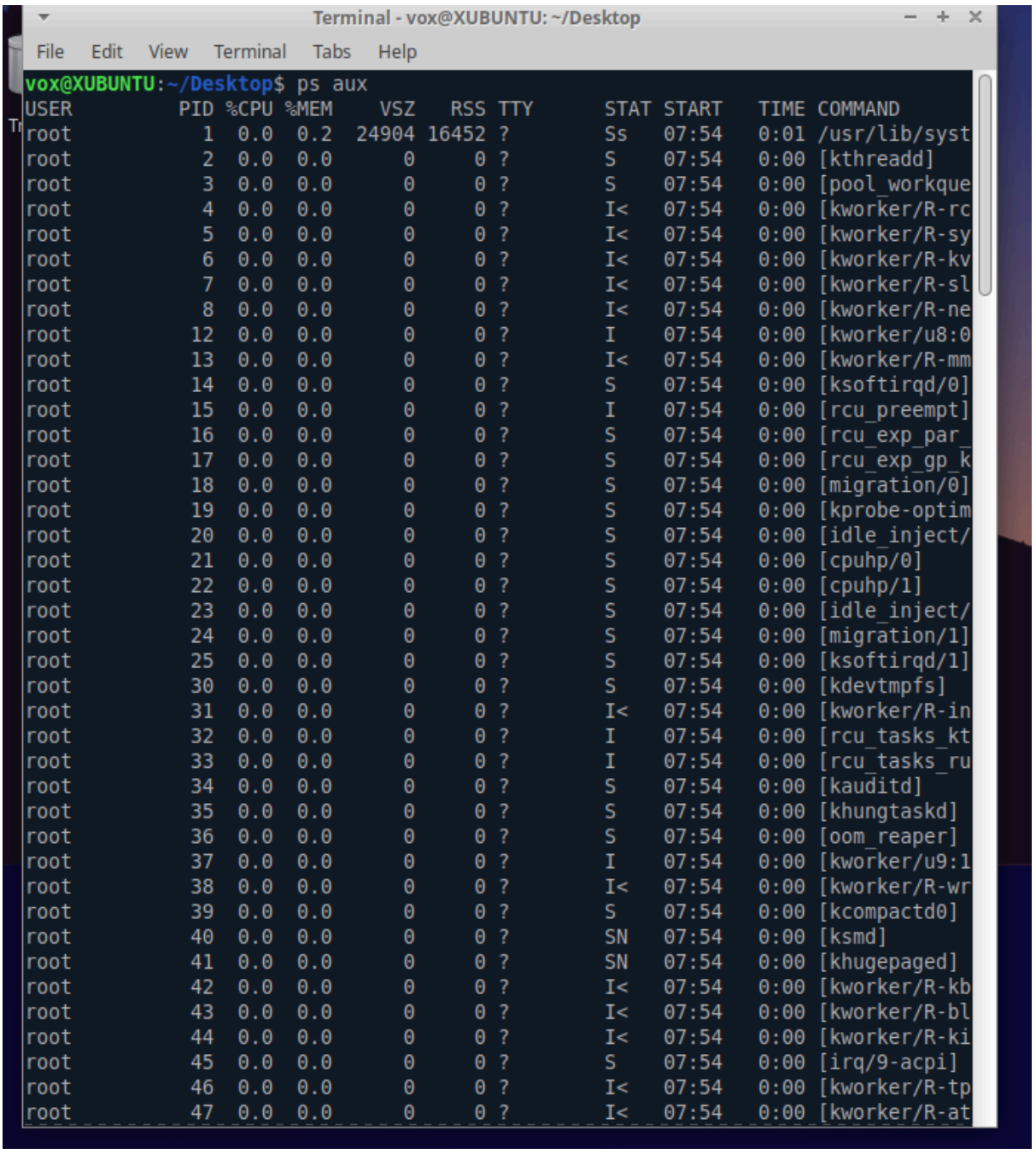

Terminal - vox@XUBUNTU: ...
Terminal - vox@XUBUNTU: ~/Desktop
File Edit View Terminal Tabs Help
Press H or ? for help
25.0Kb
37.5Kb
50.0Kb
62.5Kb
XUBUNTU => 250.red-80-58-61.staticip.r 648b 324b 324b
<= 1.12Kb 576b 576b
XUBUNTU => 114.pool84-77-195.dynamic.o 304b 152b 152b
<= 304b 152b 152b
TX: cum: 238B peak: 952b rates: 952b 476b 476b
RX: 364B 1.42Kb 1.42Kb 728b 728b
TOTAL: 602B 2.35Kb 2.35Kb 1.18Kb 1.18Kb

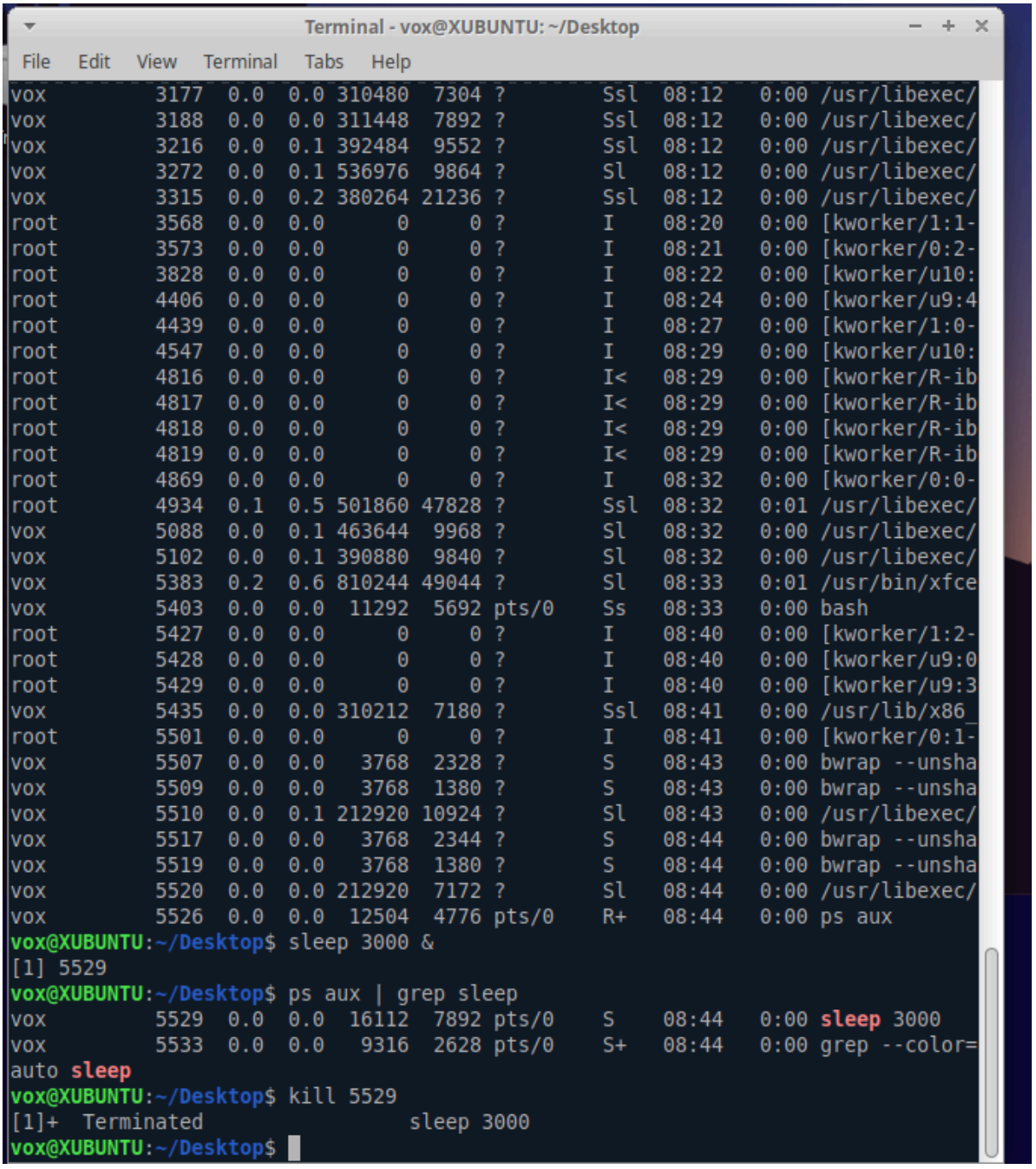
(F) Muestra estadísticas de CPU/memoria y E/S con vmstat e iostat.
Pista: se ejecutan con un intervalo en segundos.
Utilidad: análisis de rendimiento continuo.



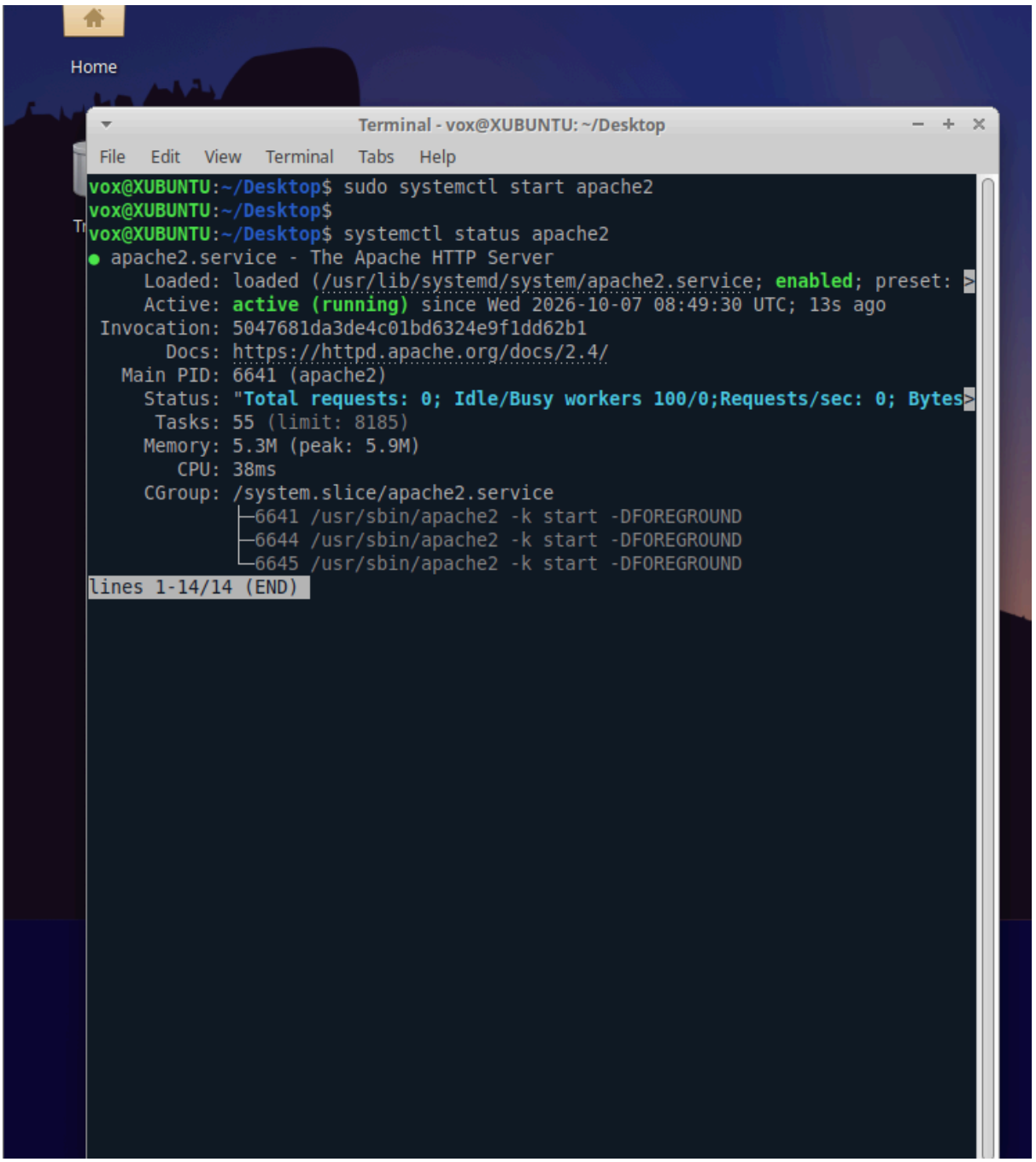


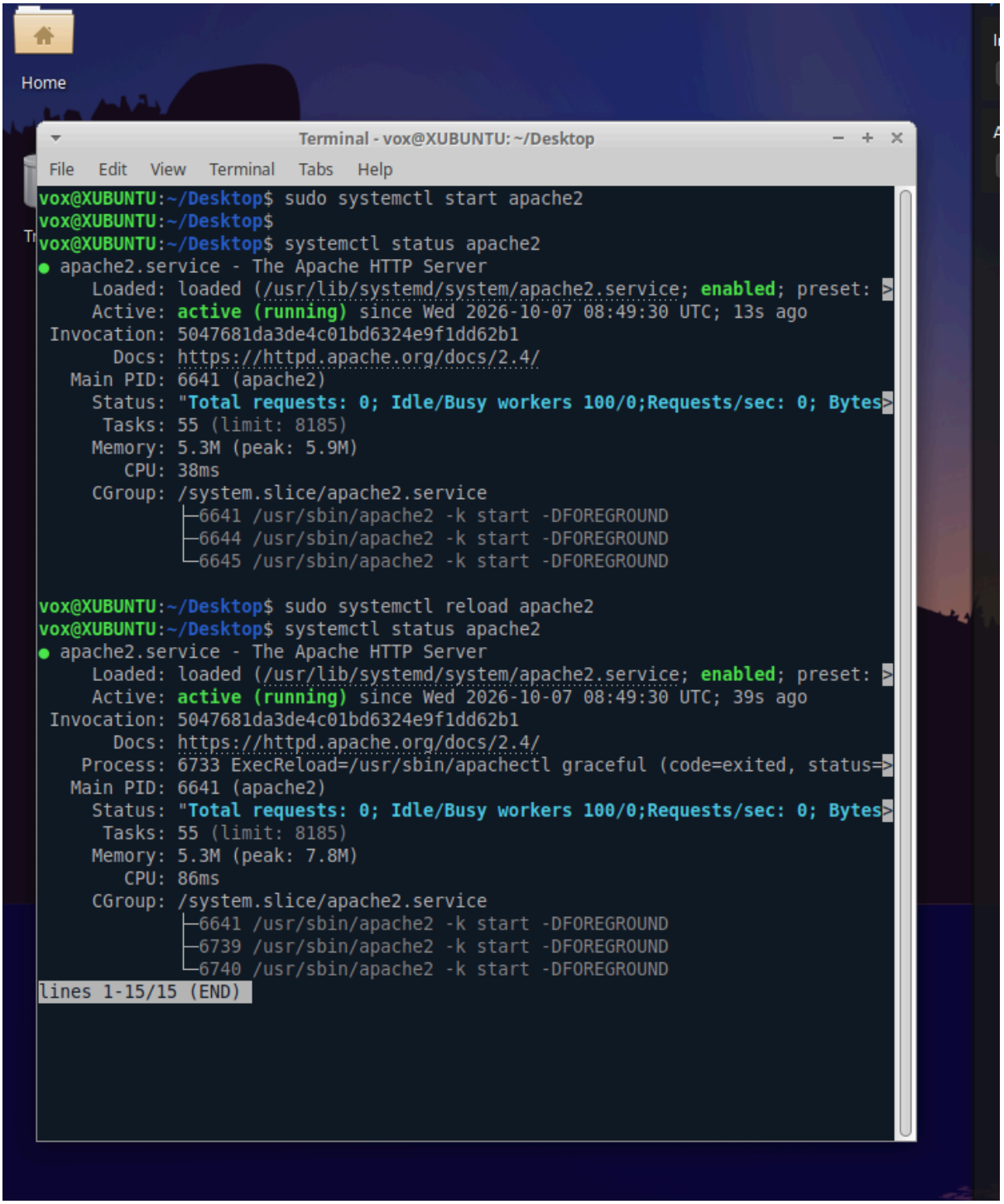
(F) Lista todos los procesos y mata uno por PID.
Pista: ps aux, kill, kill -9.
Utilidad: liberar un recurso bloqueado.

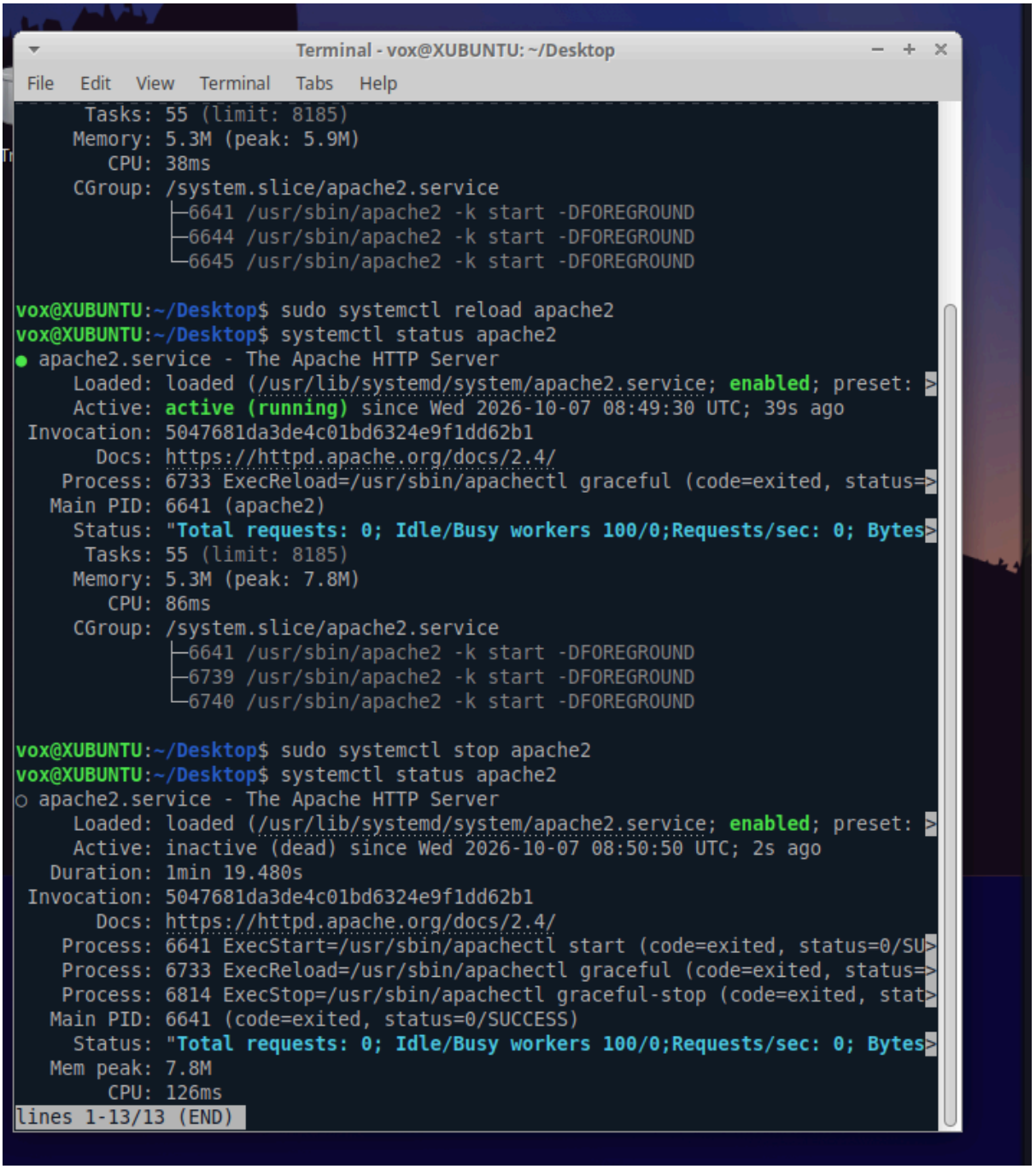

Terminal - vox@XUBUNTU: ~/Desktop
File Edit View Terminal Tabs Help
vox@XUBUNTU:~/Desktop$ ps aux
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 0.0 0.2 24904 16452 ? Ss 07:54 0:01 /usr/lib/syst
root 2 0.0 0.0 0 0 ? S 07:54 0:00 [kthreadd]
root 3 0.0 0.0 0 0 ? S 07:54 0:00 [pool_workque
root 4 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-rc
root 5 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-sy
root 6 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-kv
root 7 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-sl
root 8 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-ne
root 12 0.0 0.0 0 0 ? I 07:54 0:00 [kworker/u8:0
root 13 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-mm
root 14 0.0 0.0 0 0 ? S 07:54 0:00 [ksoftirqd/0]
root 15 0.0 0.0 0 0 ? I 07:54 0:00 [rcu_preempt]
root 16 0.0 0.0 0 0 ? S 07:54 0:00 [rcu_exp_par_
root 17 0.0 0.0 0 0 ? S 07:54 0:00 [rcu_exp_gp_k
root 18 0.0 0.0 0 0 ? S 07:54 0:00 [migration/0]
root 19 0.0 0.0 0 0 ? S 07:54 0:00 [kprobe-optim
root 20 0.0 0.0 0 0 ? S 07:54 0:00 [idle_inject/
root 21 0.0 0.0 0 0 ? S 07:54 0:00 [cpuhp/0]
root 22 0.0 0.0 0 0 ? S 07:54 0:00 [cpuhp/1]
root 23 0.0 0.0 0 0 ? S 07:54 0:00 [idle_inject/
root 24 0.0 0.0 0 0 ? S 07:54 0:00 [migration/1]
root 25 0.0 0.0 0 0 ? S 07:54 0:00 [ksoftirqd/1]
root 30 0.0 0.0 0 0 ? S 07:54 0:00 [kdevtmpfs]
root 31 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-in
root 32 0.0 0.0 0 0 ? I 07:54 0:00 [rcu_tasks_kt
root 33 0.0 0.0 0 0 ? I 07:54 0:00 [rcu_tasks_ru
root 34 0.0 0.0 0 0 ? S 07:54 0:00 [kauditd]
root 35 0.0 0.0 0 0 ? S 07:54 0:00 [khungtaskd]
root 36 0.0 0.0 0 0 ? S 07:54 0:00 [oom_reaper]
root 37 0.0 0.0 0 0 ? I 07:54 0:00 [kworker/u9:1
root 38 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-wr
root 39 0.0 0.0 0 0 ? S 07:54 0:00 [kcompactd0]
root 40 0.0 0.0 0 0 ? SN 07:54 0:00 [ksmd]
root 41 0.0 0.0 0 0 ? SN 07:54 0:00 [khugepaged]
root 42 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-kb
root 43 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-bl
root 44 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-ki
root 45 0.0 0.0 0 0 ? S 07:54 0:00 [irq/9-acpi]
root 46 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-tp
root 47 0.0 0.0 0 0 ? I< 07:54 0:00 [kworker/R-at

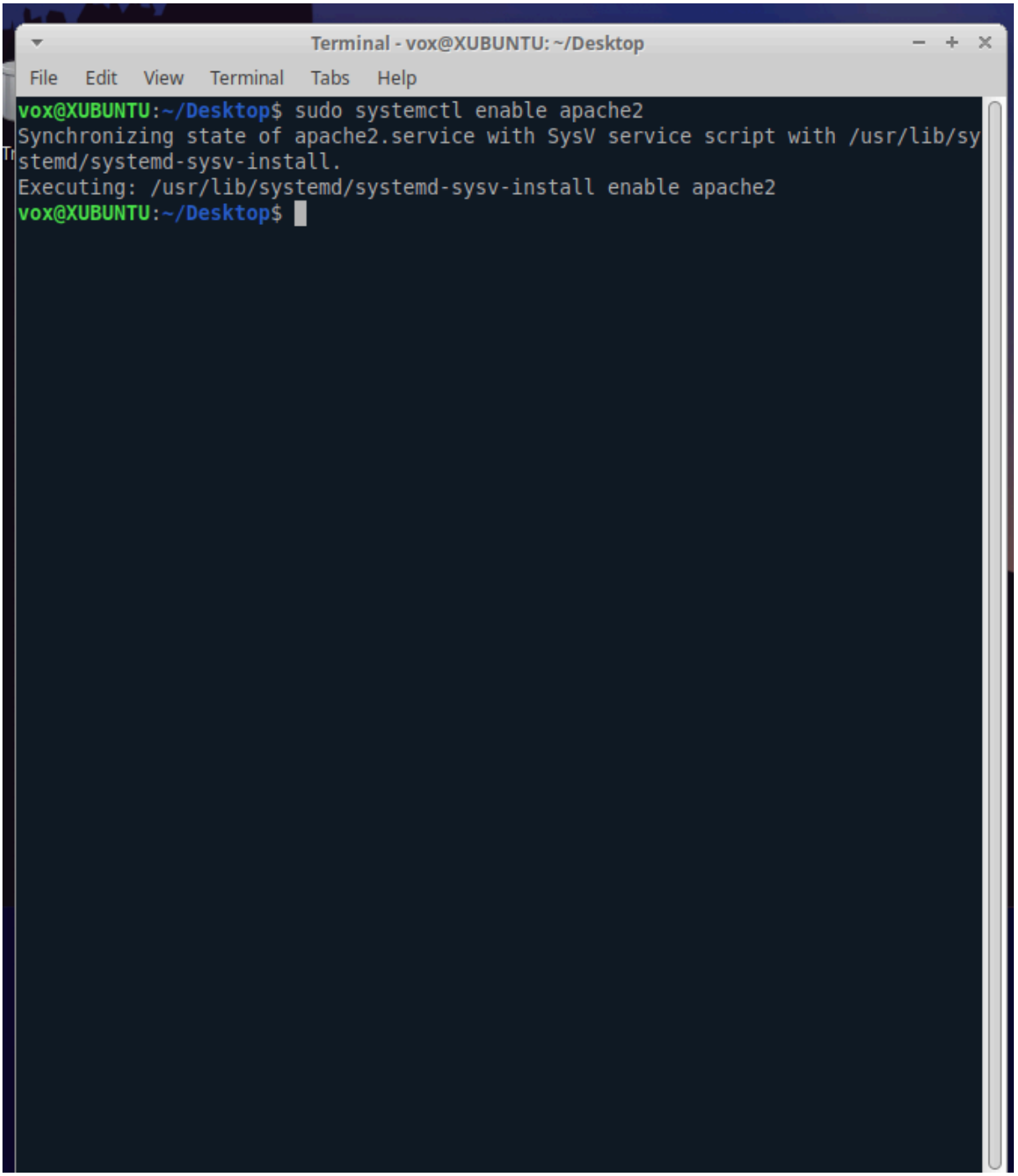


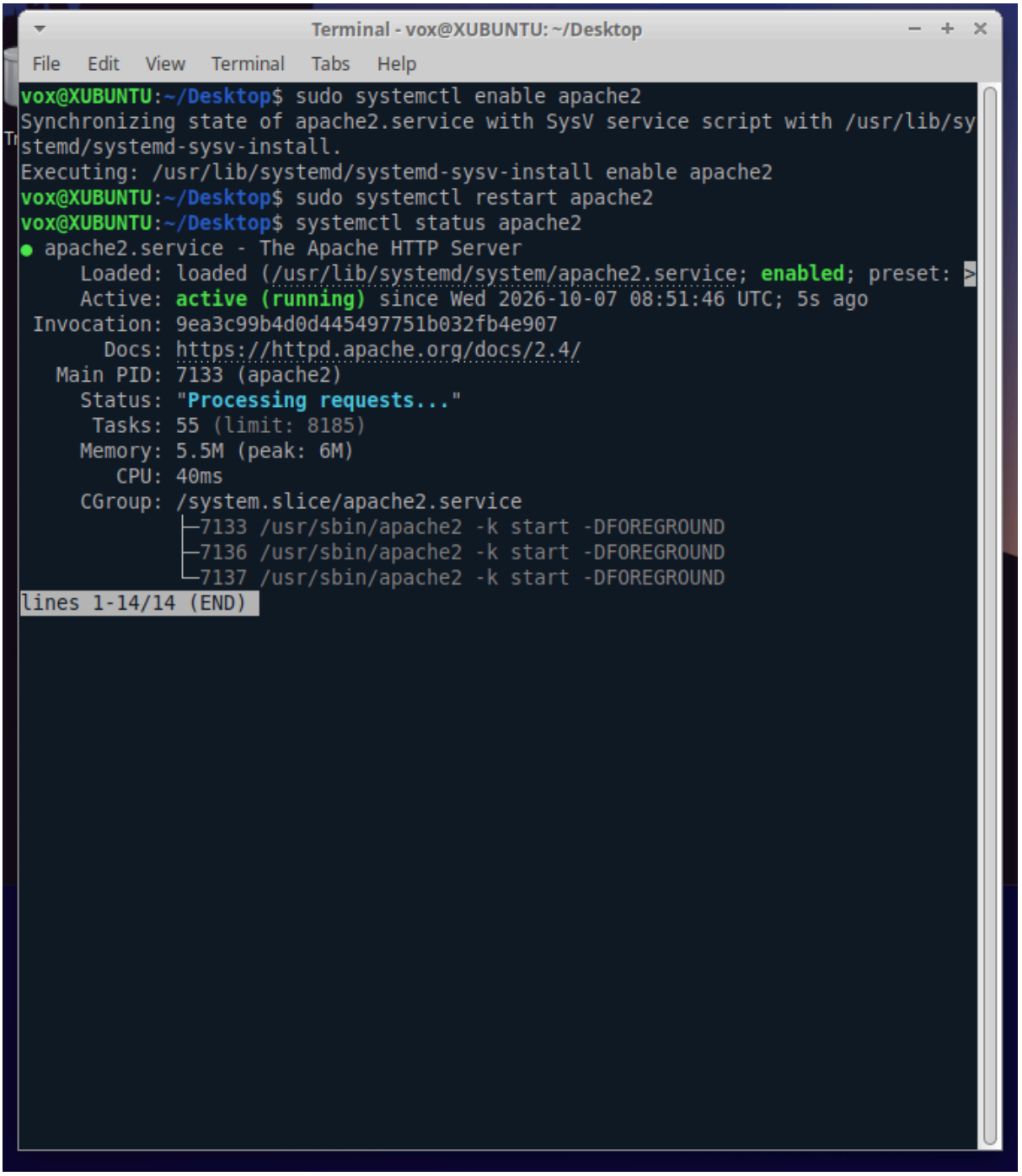
(F) Gestiona el servicio apache2: inicia, detén, reinicia, recarga y habilita al arranque.
Pista: systemctl start/stop/restart/reload/enable/status.
Utilidad: mantener servicios web en producción.


Home
Terminal - vox@XUBUNTU: ~/Desktop
File Edit View Terminal Tabs Help
vox@XUBUNTU:~/Desktop$ sudo systemctl start apache2
vox@XUBUNTU:~/Desktop$
vox@XUBUNTU:~/Desktop$ systemctl status apache2
● apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
Active: active (running) since Wed 2026-10-07 08:49:30 UTC; 13s ago
Invocation: 5047681da3de4c01bd6324e9f1dd62b1
Docs: https://httpd.apache.org/docs/2.4/
Main PID: 6641 (apache2)
Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes
Tasks: 55 (limit: 8185)
Memory: 5.3M (peak: 5.9M)
CPU: 38ms
CGroup: /system.slice/apache2.service
├─6641 /usr/sbin/apache2 -k start -DFOREGROUND
├─6644 /usr/sbin/apache2 -k start -DFOREGROUND
└─6645 /usr/sbin/apache2 -k start -DFOREGROUND
lines 1-14/14 (END)


Home
Terminal - vox@XUBUNTU: ~/Desktop
File Edit View Terminal Tabs Help
vox@XUBUNTU:~/Desktop$ sudo systemctl start apache2
vox@XUBUNTU:~/Desktop$
vox@XUBUNTU:~/Desktop$ systemctl status apache2
● apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
Active: active (running) since Wed 2026-10-07 08:49:30 UTC; 13s ago
Invocation: 5047681da3de4c01bd6324e9f1dd62b1
Docs: https://httpd.apache.org/docs/2.4/
Main PID: 6641 (apache2)
Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes
Tasks: 55 (limit: 8185)
Memory: 5.3M (peak: 5.9M)
CPU: 38ms
CGroup: /system.slice/apache2.service
├─6641 /usr/sbin/apache2 -k start -DFOREGROUND
├─6644 /usr/sbin/apache2 -k start -DFOREGROUND
└─6645 /usr/sbin/apache2 -k start -DFOREGROUND
vox@XUBUNTU:~/Desktop$ sudo systemctl reload apache2
vox@XUBUNTU:~/Desktop$ systemctl status apache2
● apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
Active: active (running) since Wed 2026-10-07 08:49:30 UTC; 39s ago
Invocation: 5047681da3de4c01bd6324e9f1dd62b1
Docs: https://httpd.apache.org/docs/2.4/
Process: 6733 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=
Main PID: 6641 (apache2)
Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes
Tasks: 55 (limit: 8185)
Memory: 5.3M (peak: 7.8M)
CPU: 86ms
CGroup: /system.slice/apache2.service
├─6641 /usr/sbin/apache2 -k start -DFOREGROUND
├─6739 /usr/sbin/apache2 -k start -DFOREGROUND
└─6740 /usr/sbin/apache2 -k start -DFOREGROUND
lines 1-15/15 (END)


Terminal - vox@XUBUNTU: ~/Desktop
File Edit View Terminal Tabs Help
Tasks: 55 (limit: 8185)
Memory: 5.3M (peak: 5.9M)
CPU: 38ms
CGroup: /system.slice/apache2.service
├─6641 /usr/sbin/apache2 -k start -DFOREGROUND
├─6644 /usr/sbin/apache2 -k start -DFOREGROUND
└─6645 /usr/sbin/apache2 -k start -DFOREGROUND
vox@XUBUNTU:~/Desktop$ sudo systemctl reload apache2
vox@XUBUNTU:~/Desktop$ systemctl status apache2
● apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
Active: active (running) since Wed 2026-10-07 08:49:30 UTC; 39s ago
Invocation: 5047681da3de4c01bd6324e9f1dd62b1
Docs: https://httpd.apache.org/docs/2.4/
Process: 6733 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=
Main PID: 6641 (apache2)
Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes
Tasks: 55 (limit: 8185)
Memory: 5.3M (peak: 7.8M)
CPU: 86ms
CGroup: /system.slice/apache2.service
├─6641 /usr/sbin/apache2 -k start -DFOREGROUND
├─6739 /usr/sbin/apache2 -k start -DFOREGROUND
└─6740 /usr/sbin/apache2 -k start -DFOREGROUND
vox@XUBUNTU:~/Desktop$ sudo systemctl stop apache2
vox@XUBUNTU:~/Desktop$ systemctl status apache2
○ apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
Active: inactive (dead) since Wed 2026-10-07 08:50:50 UTC; 2s ago
Duration: 1min 19.480s
Invocation: 5047681da3de4c01bd6324e9f1dd62b1
Docs: https://httpd.apache.org/docs/2.4/
Process: 6641 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SU
Process: 6733 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=
Process: 6814 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, stat
Main PID: 6641 (code=exited, status=0/SUCCESS)
Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes
Mem peak: 7.8M
CPU: 126ms
lines 1-13/13 (END)


Terminal - vox@XUBUNTU: ~/Desktop
File Edit View Terminal Tabs Help
vox@XUBUNTU:~/Desktop$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/sy
stemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
vox@XUBUNTU:~/Desktop$


Terminal - vox@XUBUNTU: ~/Desktop
File Edit View Terminal Tabs Help
vox@XUBUNTU:~/Desktop$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
vox@XUBUNTU:~/Desktop$ sudo systemctl restart apache2
vox@XUBUNTU:~/Desktop$ systemctl status apache2
● apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
Active: active (running) since Wed 2026-10-07 08:51:46 UTC; 5s ago
Invocation: 9ea3c99b4d0d445497751b032fb4e907
Docs: https://httpd.apache.org/docs/2.4/
Main PID: 7133 (apache2)
Status: "Processing requests..."
Tasks: 55 (limit: 8185)
Memory: 5.5M (peak: 6M)
CPU: 40ms
CGroup: /system.slice/apache2.service
├─7133 /usr/sbin/apache2 -k start -DFOREGROUND
├─7136 /usr/sbin/apache2 -k start -DFOREGROUND
└─7137 /usr/sbin/apache2 -k start -DFOREGROUND
lines 1-14/14 (END)